### Activity 1

(i) Observe the external morphology of leech specimen in your biology laboratory.

(ii) Can you find leeches in your locality?

(iii) In which geographical areas are leeches found more predominantly in India?

(v) **Female genital pore:** It lies mid-ventrally between second and third annuli of 11th segment.

## 13.1.3 Divisions of the Body

The body of leech is divided into six regions.

| Region | Segments |
|---|---|
| Cephalic region | 1st - 5th |
| Pre-clitellar region | 6th, 7th and 8th |
| Clitellar region | 9th, 10th and 11th |
| Middle region | 12th - 22nd |
| Caudal region | 23rd - 26th |
| Posterior sucker | 27th - 33rd |

## 13.1.4 Body wall

Body wall of leech includes five layers: (i) **cuticle** (outermost layer) (ii) **epidermis** which lies below the cuticle (iii) **dermis** which lies below the epidermis formed of connective tissue (iv) **muscular layer** formed of circular and longitudinal muscles (v) **botryoidal tissue** lies beneath longitudinal muscles and fills the entire coelom around the gut.

## 13.1.5 Locomotion

Locomotion in leech takes place by (i) looping or crawling movement (ii) Swimming movement.

### (i) Looping or Crawling movement

This type of movement is brought about by the contraction and relaxation of muscles. The two suckers serve for attachment during movement on a substratum.

### (ii) Swimming movement

Leeches swim very actively and perform undulating movements in water.

## 13.1.6 Digestive System

The digestive system includes the long alimentary canal and the digestive glands.

### Alimentary Canal

The alimentary canal of leech is a straight tube running from the mouth to the anus. **Mouth** is a **triradiate** aperture situated in the middle of the anterior sucker that leads into the small buccal cavity. The wall of the buccal cavity bears three jaws with single row of minute teeth. The jaws are provided with **papillae** which bear the openings of salivary glands. Mouth and buccal cavity occupy the first five segments.

The buccal cavity leads into muscular **pharynx.** It is surrounded by salivary glands. The secretion of saliva contains **hirudin** which prevents the coagulation of blood. Pharynx leads into crop through a short and narrow **oesophagus.**

**Crop** is the largest portion of the alimentary canal. It is divided into a series of 10 chambers. The chambers communicate with one another through circular apertures surrounded by **sphincters.** A pair of lateral, backwardly directed caecae arises as blind outgrowth from each chamber known as **caeca** or **diverticula.** Crop and its diverticula can store large amount of blood which can be slowly digested.

The last chamber of crop opens into stomach. The stomach leads into **intestine** which is a small straight tube that opens into **rectum.** The rectum opens to the exterior by anus.



10th_Science Unit-13.indd 189 06-12-2021 18:39:57